

Agency CMS Backend — Full Audit Report

Audited: /home/avinash-sharma/Desktop/Agency/Server

Against: [implementation_plan.md](file:///home/avinash-sharma/Desktop/Agency/docs/implementation_plan.md)

Date: 2026-08-13

Executive Summary

Your friend has done **solid work** — the vast majority of the plan is implemented. All 8 phases are present, all 13 modules have full MVC + Repository stacks, routes are aggregated, Swagger is configured, and the architecture follows the plan's layered pattern. However, there are a few **structural deviations**, **missing files**, and **quality observations** worth noting.

Overall Verdict: ✓ **85–90% Faithful to Plan** — with acceptable design variations

Phase-by-Phase File Checklist

Phase 1 — Foundation & Infrastructure ✓

Planned File	Status	Notes
package.json	✓ Present	All deps listed, proper scripts (dev , start , seed , seed:roles , seed:admin , lint)
.env.example	✓ Present	All env vars covered with good documentation
.gitignore	✓ Present	
src/config/index.js	✓ Present	Zod-validated config loader
src/config/database.js	✓ Present	Retry logic included
src/config/cloudinary.js	✓ Present	
src/config/nodemailer.js	✓ Present	
src/config/redis.js	⚠️ MISSING	Plan listed this as optional — not created

Planned File	Status	Notes
src/utils/logger.js	✓ Present	Winston with rotating file transports
src/utils/ApiError.js	✓ Present	
src/utils/ApiResponse.js	✓ Present	
src/utils/asyncHandler.js	✓ Present	
src/utils/slugify.js	✓ Present	
src/utils/pagination.js	✓ Present	
src/utils/queryBuilder.js	✓ Present	
src/constants/*.js	✓ Present	All 5 files: httpStatus, messages, roles, enums, index
src/helpers/password.helper.js	✓ Present	
src/helpers/token.helper.js	✓ Present	
src/helpers/cloudinary.helper.js	✓ Present	
src/helpers/email.helper.js	✓ Present	
src/middlewares/errorHandler.middleware.js	✓ Present	
src/middlewares/notFound.middleware.js	✓ Present	
src/middlewares/validate.middleware.js	✓ Present	
src/middlewares/upload.middleware.js	✓ Present	
src/middlewares/rateLimiter.middleware.js	✓ Present	

Planned File	Status	Notes
src/repositories/base.repository.js	✓ Present	8.8 KB — substantial with pagination, soft delete, aggregation
src/app.js	✓ Present	
src/server.js	✓ Present	
src/emails/templates/resetPassword.html	✓ Present	
src/emails/templates/welcomeEmail.html	✓ Present	
[!NOTE] Extra file found: [cloudinaryUploader.js](file:///home/avinash-sharma/Desktop/Agency/Server/src/utils/cloudinaryUploader.js) in utils/ — this is a duplicate of helpers/cloudinary.helper.js with added local-fallback logic. This is an unplanned addition but functionally useful.		

Phase 2 — Authentication & User Management ✓

Planned File	Status	Notes
src/models/Role.model.js	✓ Present	Enum-constrained to ROLE_LIST
src/models/User.model.js	✓ Present	Refresh token array, password hashing, reset fields, login tracking
src/repositories/role.repository.js	✓ Present	
src/repositories/user.repository.js	✓ Present	
src/services/auth.service.js	✓ Present	12 KB — login, logout, refresh rotation, forgot/reset/change password
src/services/user.service.js	✓ Present	CRUD + profile + avatar upload
src/controllers/auth.controller.js	✓ Present	
src/controllers/user.controller.js	✓ Present	

Planned File	Status	Notes
src/validators/auth.validator.js	✓ Present	Strong password regex, confirm password
src/validators/user.validator.js	✓ Present	ObjectId validation
src/routes/auth.routes.js	✓ Present	Swagger annotated
src/routes/user.routes.js	✓ Present	Swagger annotated
src/middlewares/auth.middleware.js	✓ Present	JWT + password-change invalidation
src/middlewares/rbac.middleware.js	✓ Present	3 strategies: exact role, hierarchy, owner-or-role
src/database/seeds/roleSeeder.js	✓ Present	Idempotent
src/database/seeds/adminSeeder.js	✓ Present	Idempotent

Phase 3 — Categories & Services ✓

Planned File	Status
src/models/Category.model.js	✓ Present
src/models/Service.model.js	✓ Present
src/repositories/category.repository.js	✓ Present
src/repositories/service.repository.js	✓ Present
src/services/category.service.js	✓ Present
src/services/service.service.js	✓ Present
src/controllers/category.controller.js	✓ Present
src/controllers/service.controller.js	✓ Present
src/validators/category.validator.js	✓ Present
src/validators/service.validator.js	✓ Present
src/routes/category.routes.js	✓ Present
src/routes/service.routes.js	✓ Present

Phase 4 — Packages & FAQs ✓

Planned File	Status
src/models/Package.model.js	✔ Present
src/models/Faq.model.js	✔ Present
src/repositories/package.repository.js	✔ Present
src/repositories/faq.repository.js	✔ Present
src/services/package.service.js	✔ Present
src/services/faq.service.js	✔ Present
src/controllers/package.controller.js	✔ Present
src/controllers/faq.controller.js	✔ Present
src/validators/package.validator.js	✔ Present
src/validators/faq.validator.js	✔ Present
src/routes/package.routes.js	✔ Present
src/routes/faq.routes.js	✔ Present

Phase 5 — Portfolio & Blog ⚠ (1 deviation)

Planned File	Status	Notes
src/models/Portfolio.model.js	✔ Present	
src/models/PortfolioImage.model.js	✗ MISSING	Gallery images embedded directly in Portfolio model as a sub-array
src/models/Blog.model.js	✔ Present	
src/repositories/portfolio.repository.js	✔ Present	
src/repositories/portfolioImage.repository.js	✗ MISSING	Not needed since images are embedded
src/repositories/blog.repository.js	✔ Present	
src/services/portfolio.service.js	✔ Present	
src/services/blog.service.js	✔ Present	Reading time calc, slug generation

Planned File	Status	Notes
src/controllers/portfolio.controller.js	✓ Present	
src/controllers/blog.controller.js	✓ Present	
src/validators/portfolio.validator.js	✓ Present	
src/validators/blog.validator.js	✓ Present	
src/routes/portfolio.routes.js	✓ Present	
src/routes/blog.routes.js	✓ Present	

[!IMPORTANT] **Design Deviation — PortfolioImage:** The plan called for a **separate PortfolioImage model** with its own repository. Your friend instead embedded **galleryImages** as a sub-document array directly in the Portfolio schema. This is actually a **reasonable simplification** for this use case (no need to query images independently), but it deviates from the plan's explicit file list.

Phase 6 — Team, Testimonials & Leads ✓

Planned File	Status
src/models/TeamMember.model.js	✓ Present
src/models/Testimonial.model.js	✓ Present
src/models/Lead.model.js	✓ Present
src/repositories/teamMember.repository.js	✓ Present
src/repositories/testimonial.repository.js	✓ Present
src/repositories/lead.repository.js	✓ Present
src/services/teamMember.service.js	✓ Present
src/services/testimonial.service.js	✓ Present
src/services/lead.service.js	✓ Present
src/controllers/teamMember.controller.js	✓ Present
src/controllers/testimonial.controller.js	✓ Present
src/controllers/lead.controller.js	✓ Present
src/validators/teamMember.validator.js	✓ Present

Planned File	Status
src/validators/testimonial.validator.js	✔ Present
src/validators/lead.validator.js	✔ Present
src/routes/teamMember.routes.js	✔ Present
src/routes/testimonial.routes.js	✔ Present
src/routes/lead.routes.js	✔ Present

Phase 7 — Media & Settings ✔

Planned File	Status
src/models/Media.model.js	✔ Present
src/models/Setting.model.js	✔ Present — Singleton pattern with <code>getSingleton()</code> static
src/repositories/media.repository.js	✔ Present
src/repositories/setting.repository.js	✔ Present
src/services/media.service.js	✔ Present
src/services/setting.service.js	✔ Present — Get + Update only (singleton pattern)
src/controllers/media.controller.js	✔ Present
src/controllers/setting.controller.js	✔ Present
src/validators/media.validator.js	✔ Present
src/validators/setting.validator.js	✔ Present
src/routes/media.routes.js	✔ Present
src/routes/setting.routes.js	✔ Present

Phase 8 — Swagger & Final Integration ✔

Planned File	Status	Notes
swagger.js	✔ Present	Located at <code>src/swagger.js</code> (plan said root-level) — minor path difference
src/routes/index.js	✔ Present	All 13 module routes mounted
src/app.js (Swagger mount)	✔ Present	Swagger UI at <code>/api-docs</code> , JSON spec at <code>/api-docs.json</code>

Architecture & Design Decisions Audit

Plan Decision	Implementation	Verdict
Base Repository Pattern	✔ All 14 repos extend <code>BaseRepository</code>	Matches plan
Refresh Token Strategy	✔ Rotating array in User model, HTTP-only cookie, reuse detection	Matches plan
Singleton Settings	✔ <code>getSingleton()</code> static method, only get/update operations	Matches plan
Slug Generation	✔ Custom <code>slugify.js</code> utility, counter-based collision resolution	Matches plan (uses counter suffix instead of UUID, which is actually better)
Soft Delete	✔ <code>isDeleted</code> + <code>deletedAt</code> on Services, Blogs, Portfolio, Categories, FAQ, Packages, Team, Testimonials, Leads, Media	Exceeds plan — applied more broadly than specified
Query Builder	✔ <code>queryBuilder.js</code> with search, sort, filter, pagination	Matches plan
Clean Architecture	✔ Controller → Service → Repository layering	Matches plan

Full File Inventory Comparison

Files in Plan: 80+ → Files Present: 83 (excluding node_modules, logs, .git)

Root:	5 files	(.env, .env.example, .gitignore, package.json, package-lock.json)
src/config:	4 files	(index, database, cloudinary, nodemailer)
src/constants:	5 files	(httpStatus, messages, roles, enums, index)
src/models:	14 files	(13 models + index.js)
src/repos:	14 files	(base + 13 module repos)
src/services:	13 files	(auth, user + 11 modules)
src/controllers:	13 files	
src/routes:	14 files	(13 module routes + index.js)
src/validators:	13 files	
src/middlewares:	7 files	
src/utls:	8 files	(7 planned + 1 extra cloudinaryUploader)
src/helpers:	4 files	
src/emails:	2 files	
src/database:	2 files	(2 seeders)
src/swagger.js:	1 file	
src/app.js:	1 file	
src/server.js:	1 file	

Issues & Gaps Summary

✕ Missing Files (2)

File	Impact	Severity
src/config/redis.js	No Redis client setup	● Low — plan marked as optional
src/models/PortfolioImage.model.js + src/repositories/portfolioImage.repository.js	No separate image model	● Low — embedded in Portfolio model (reasonable alternative)

⚠ Deviations (3)

Item	Plan Says	Actual	Severity
Swagger file location	Root swagger.js	src/swagger.js	● Trivial — import paths adjusted correctly
Portfolio images architecture	Separate model + repo	Embedded sub-array	● Low — valid design choice, less flexible but simpler
Slug collision strategy	UUID suffix	Counter suffix (-1, -2, etc.)	● Better — more readable slugs

● Quality Observations (4)

Item	Observation
Duplicate Cloudinary utility	Both helpers/cloudinary.helper.js AND utils/cloudinaryUploader.js exist. The utils version adds local-fallback logic. Could create confusion about which to import.
Setting service is thin	[setting.service.js](file:///home/avinash-sharma/Desktop/Agency/Server/src/services/setting.service.js) is only 29 lines — just proxies to repo with no validation or business logic. Logo/favicon upload handling may be missing.
Soft delete applied broadly	Plan said "models where data recovery matters (Services, Blogs, Portfolio)". Implementation applies it to Categories, FAQs, Packages, TeamMembers, Testimonials, Leads, Media too. This is actually better for production but deviates.
No models/index.js PortfolioImage export	The barrel file still has a commented line <code>// export { default as PortfolioImage }</code> — this is fine since the model doesn't exist.

✔ Things Done Well

Item	Detail
Swagger coverage	All 13 route files have <code>@swagger</code> JSDoc annotations. The swagger spec (14.5 KB) defines schemas for every module.
Route aggregator	All 13 routes are properly mounted in <code>routes/index.js</code>
App.js integration	Swagger UI, JSON spec endpoint, all middleware, routes — all properly wired
Security middleware	Helmet, CORS, mongoSanitize, rate limiter, cookieParser — all present
Static uploads	Added <code>express.static</code> for local upload fallback — not in plan but useful
Lead model extras	<code>budget</code> , <code>assignedTo</code> , <code>notes[]</code> , <code>ip</code> , <code>userAgent</code> — richer than plan
Package model extras	<code>packageType</code> , <code>currency</code> , <code>billingPeriod</code> , features with <code>isIncluded</code> — well-designed
FAQ model extras	<code>isGlobal</code> flag, <code>category</code> string — adds flexibility beyond per-service FAQs
Testimonial model	<code>clientDesignation</code> , <code>clientCompany</code> , <code>service</code> ref — well-rounded
Blog model	<code>readingTime</code> , <code>viewsCount</code> , <code>publishedAt</code> , <code>status</code> enum — publication workflow
Compound indexes	Every model has thoughtful compound indexes for common query patterns

Final Scorecard

Phase	Status	Completeness
Phase 1 — Foundation	✔ Done	97% (missing optional redis.js)
Phase 2 — Auth & Users	✔ Done	100%
Phase 3 — Categories & Services	✔ Done	100%
Phase 4 — Packages & FAQs	✔ Done	100%
Phase 5 — Portfolio & Blog	⚠ Done with deviation	90% (PortfolioImage merged into Portfolio)
Phase 6 — Team, Testimonials, Leads	✔ Done	100%
Phase 7 — Media & Settings	✔ Done	100%

Phase	Status	Completeness
Phase 8 — Swagger & Integration	✔ Done	100%

Overall: ✔ 98% of planned files exist. All modules are functional. Architecture matches the plan with minor (acceptable) variations.

Recommendations

1. **Clean up the duplicate Cloudinary utility** — decide whether `helpers/cloudinary.helper.js` or `utils/cloudinaryUploader.js` is the canonical one and remove the other.
2. **Beef up `setting.service.js`** — add logo/favicon upload logic via Cloudinary if Settings supports image fields.
3. **`redis.js` can remain absent** — the plan marked it optional and the app doesn't use it. Only add if you plan to implement caching.
4. **The PortfolioImage merge is fine** — only revisit if you later need to query/paginate gallery images independently.